Koen Essers • Master Internship • March 15, 2026

Notes Week 7

a. Overshooting

I compare the single star evolution of stars with different masses for 3 different overshooting prescriptions:

1. Overshooting from Rees et al. (2024)
2. Overshooting from Temmink et al. (2023)
3. Overshooting from Temmink et al. (2023) with different $f_{ov,0}$.

Below, I note the exact inlist settings for all three cases.

a.1 Rees et al. (2024)

```
! inlist_MS
&controls
    ...
        overshoot_scheme(1) = 'step' !jermyn et al 2022
        overshoot_zone_type(1) = 'any'
        overshoot_zone_loc(1) = 'core'
        overshoot_bdy_loc(1) = 'any'
        overshoot_f(1) = 0.129
        overshoot_f0(1) = 0.0129
        overshoot_D0(1) = 0d0
        overshoot_Delta0(1) = 1d0
        overshoot_scheme(2) = 'exponential'
        overshoot_zone_type(2) = 'nonburn'
        overshoot_zone_loc(2) = 'shell'
        overshoot_bdy_loc(2) = 'any'
        overshoot_f(2) = 0.0174
        overshoot_f0(2) = 0.00174
        overshoot_D0(2) = 0d0
        overshoot_Delta0(2) = 1d0
    ...
```

```
! inlist_GB
&controls
    ...
        overshoot_scheme(1) = 'exponential'
        overshoot_zone_type(1) = 'nonburn'
        overshoot_zone_loc(1) = 'shell'
        overshoot_bdy_loc(1) = 'any'
        overshoot_f(1) = 0.0174
        overshoot_f0(1) = 0.00174
        overshoot_D0(1) = 0d0
        overshoot_Delta0(1) = 1d0
    ...
```

```
! inlist_CHeB
&controls
    ...
        predictive_mix(1) = .true.
        predictive_zone_type(1) = 'any'
        predictive_zone_loc(1) = 'core'
        predictive_bdy_loc(1) = 'any'
        predictive_superad_thresh(1) = 0.005
        predictive_avoid_reversal = 'he4'

        overshoot_scheme(2) = 'exponential'
        overshoot_zone_type(2) = 'nonburn'
        overshoot_zone_loc(2) = 'shell'
        overshoot_bdy_loc(2) = 'any'
        overshoot_f(2) = 0.0174
        overshoot_f0(2) = 0.00174
        overshoot_D0(2) = 0d0
        overshoot_Delta0(2) = 1d0
   ...
```

```
! inlist_EAGB
&controls
    ...
        overshoot_scheme(1) = 'exponential' ! IPCZ
         ↪ overshooting controls
        overshoot_zone_type(1) = 'burn_He'
        overshoot_zone_loc(1) = 'shell'
        overshoot_bdy_loc(1) = 'bottom'
        overshoot_f(1) = 0.008 ! Herwig 2008
        overshoot_f0(1) = 0.0008
        overshoot_scheme(2) = 'exponential' ! envelope
         ↪ overshooting
        overshoot_zone_type(2) = 'nonburn'
        overshoot_zone_loc(2) = 'shell'
        overshoot_bdy_loc(2) = 'any'
        overshoot_f(2) = 0.0174
        overshoot_f0(2) = 0.00174
        overshoot_scheme(3) = 'exponential' ! carbon
         ↪ burning
        overshoot_zone_type(3) = 'any'
        overshoot_zone_loc(3) = 'any'
        overshoot_bdy_loc(3) = 'bottom'
        overshoot_f(3) = 0.005
        overshoot_f0(3) = 0.0005
  ...
```

a.2 Temmink et al. (2023)

```
! inlist_common
&controls
    ...
        overshoot_scheme(1) = 'exponential'
        overshoot_zone_type(1) = 'any'
        overshoot_zone_loc(1) = 'core'
        overshoot_bdy_loc(1) = 'top'
        overshoot_f(1) = 0.016d0
        overshoot_f0(1) = 0.08d0

        overshoot_scheme(2) = 'exponential'
        overshoot_zone_type(2) = 'any'
        overshoot_zone_loc(2) = 'shell'
        overshoot_bdy_loc(2) = 'any'
        overshoot_f(2) = 0.0174d0
        overshoot_f0(2) = 0.0087d0
    ...
```

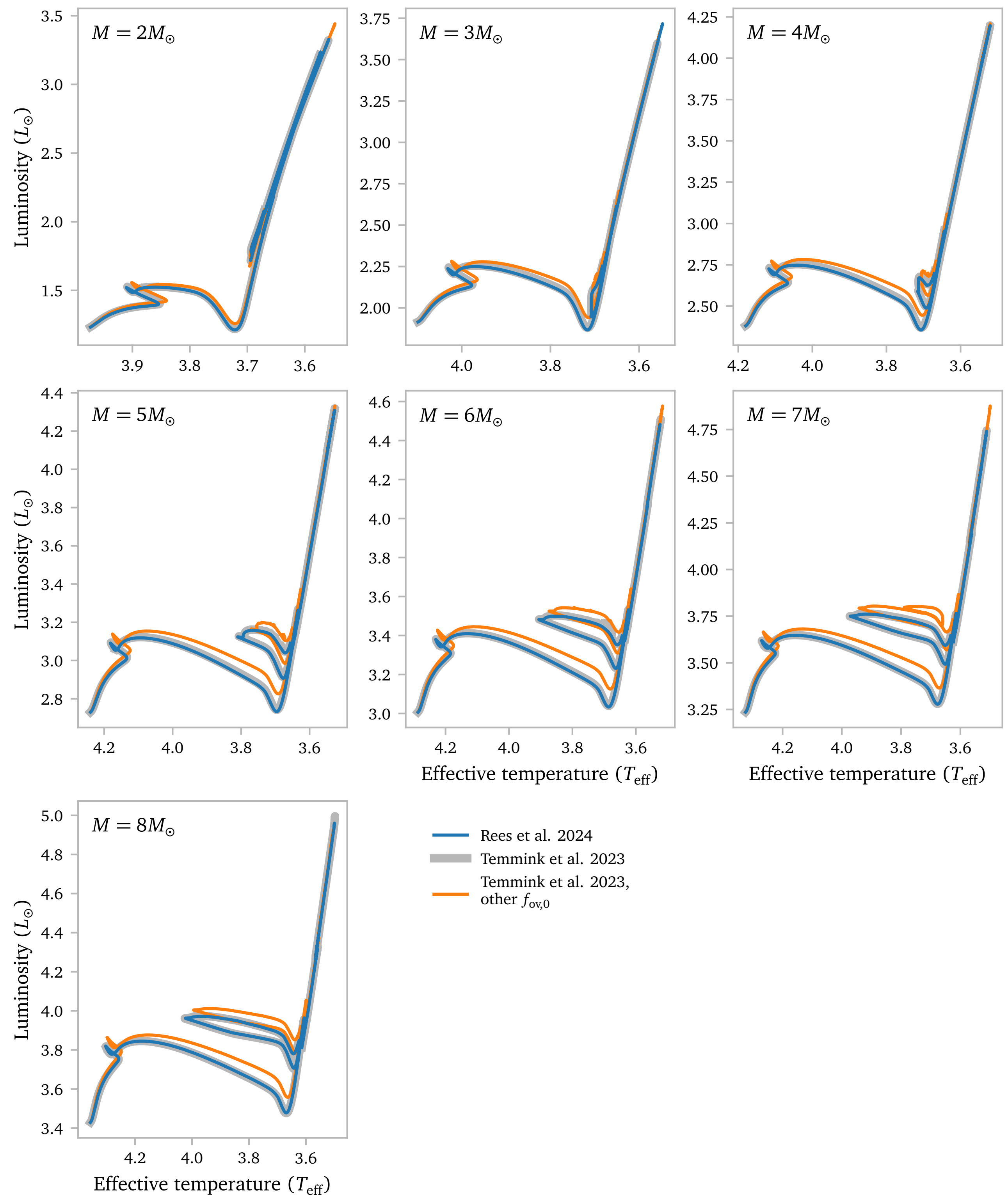
a.3 Temmink et al. (2023) other $f_{ov,0}$

```
! inlist_common
&controls
    ...
        overshoot_scheme(1) = 'exponential'
        overshoot_zone_type(1) = 'any'
        overshoot_zone_loc(1) = 'core'
        overshoot_bdy_loc(1) = 'top'
        overshoot_f(1) = 0.016d0
        overshoot_f0(1) = 0.008d0

        overshoot_scheme(2) = 'exponential'
        overshoot_zone_type(2) = 'any'
        overshoot_zone_loc(2) = 'shell'
        overshoot_bdy_loc(2) = 'any'
        overshoot_f(2) = 0.0174d0
        overshoot_f0(2) = 0.0087d0
   ...
```

a.4 Results

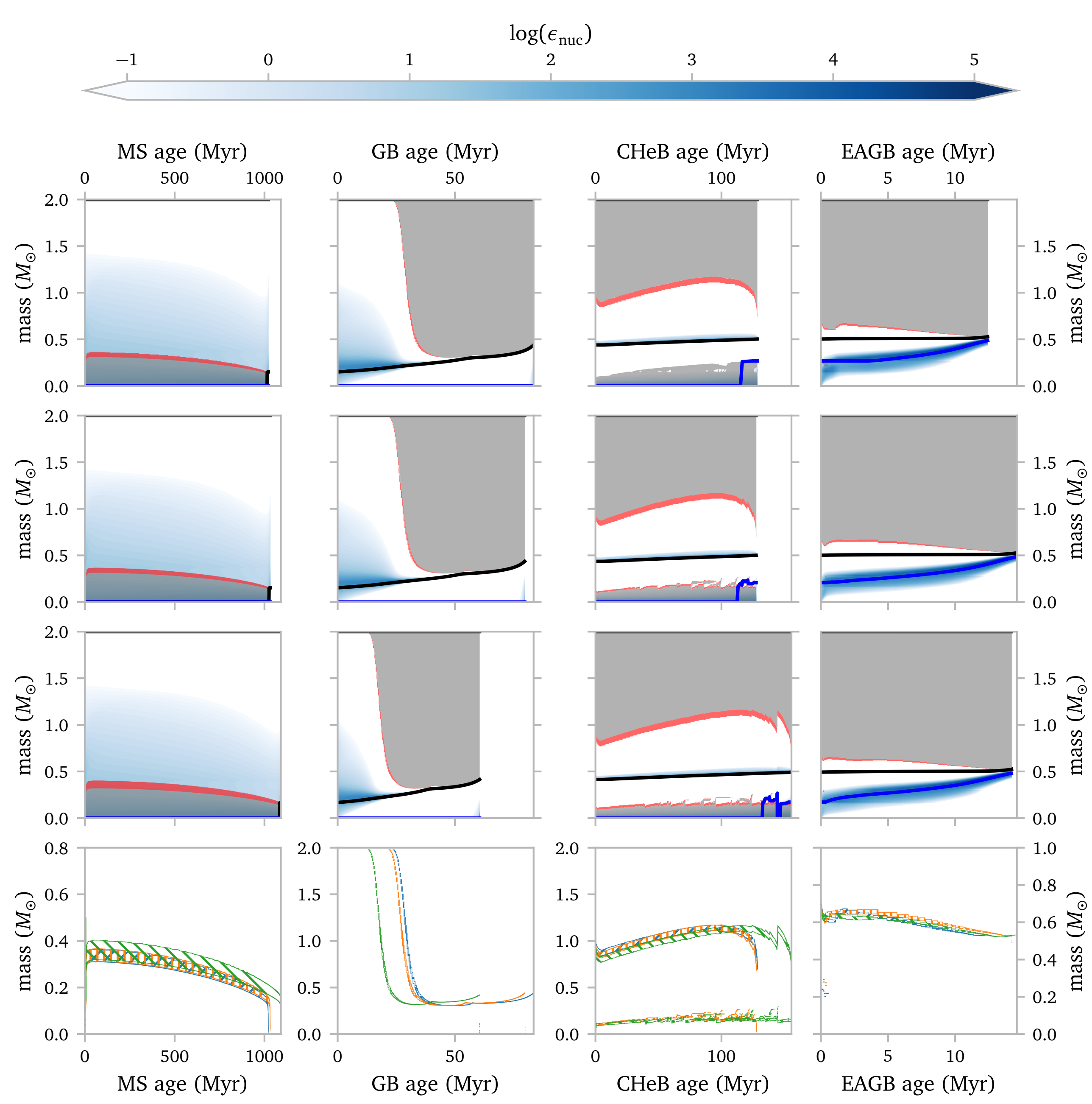
Below, I show Hertzsprung Russel diagrams for the pre-TPAGB phases. For 7 different masses, the overshooting options listed above are shown.

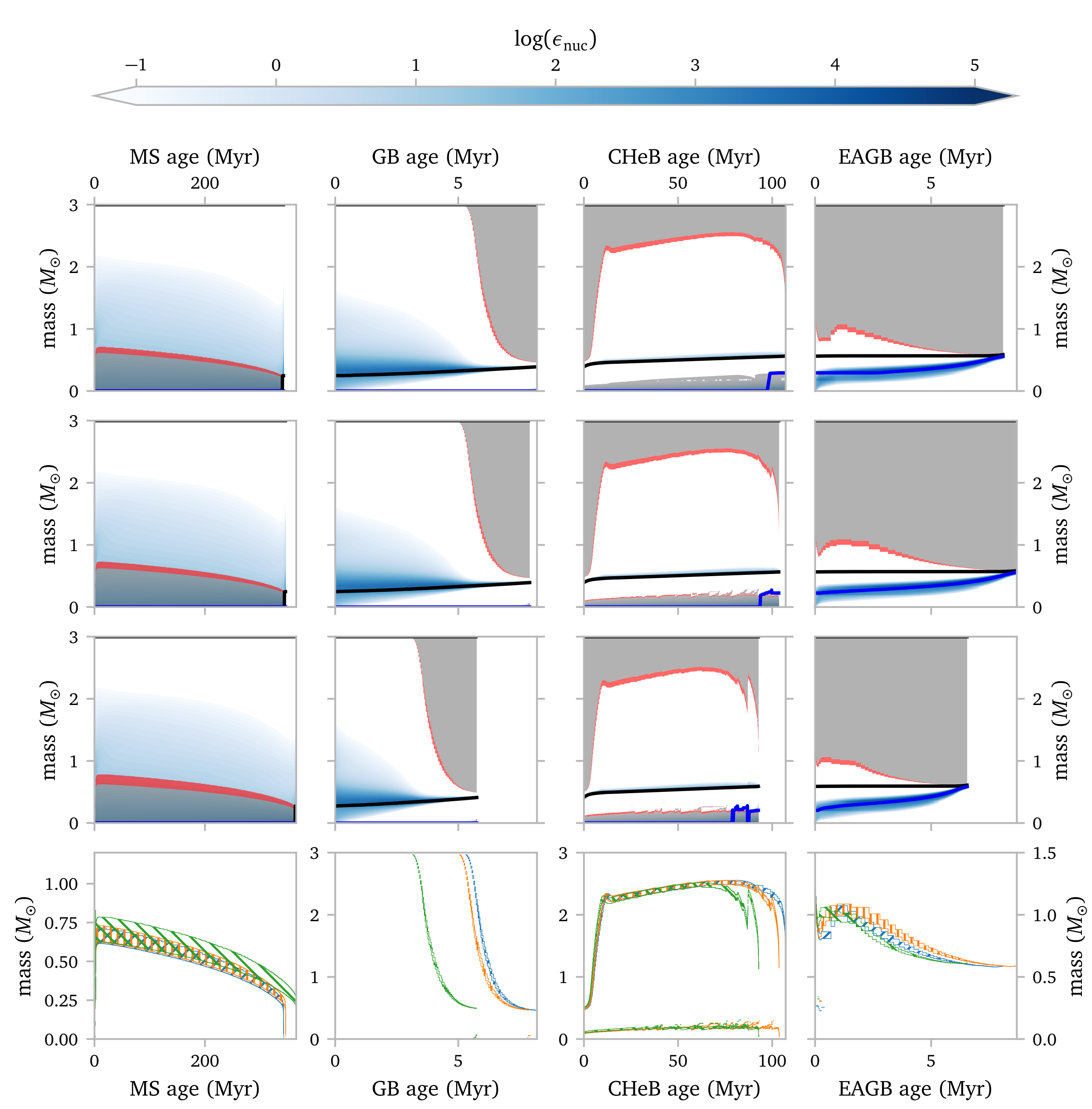


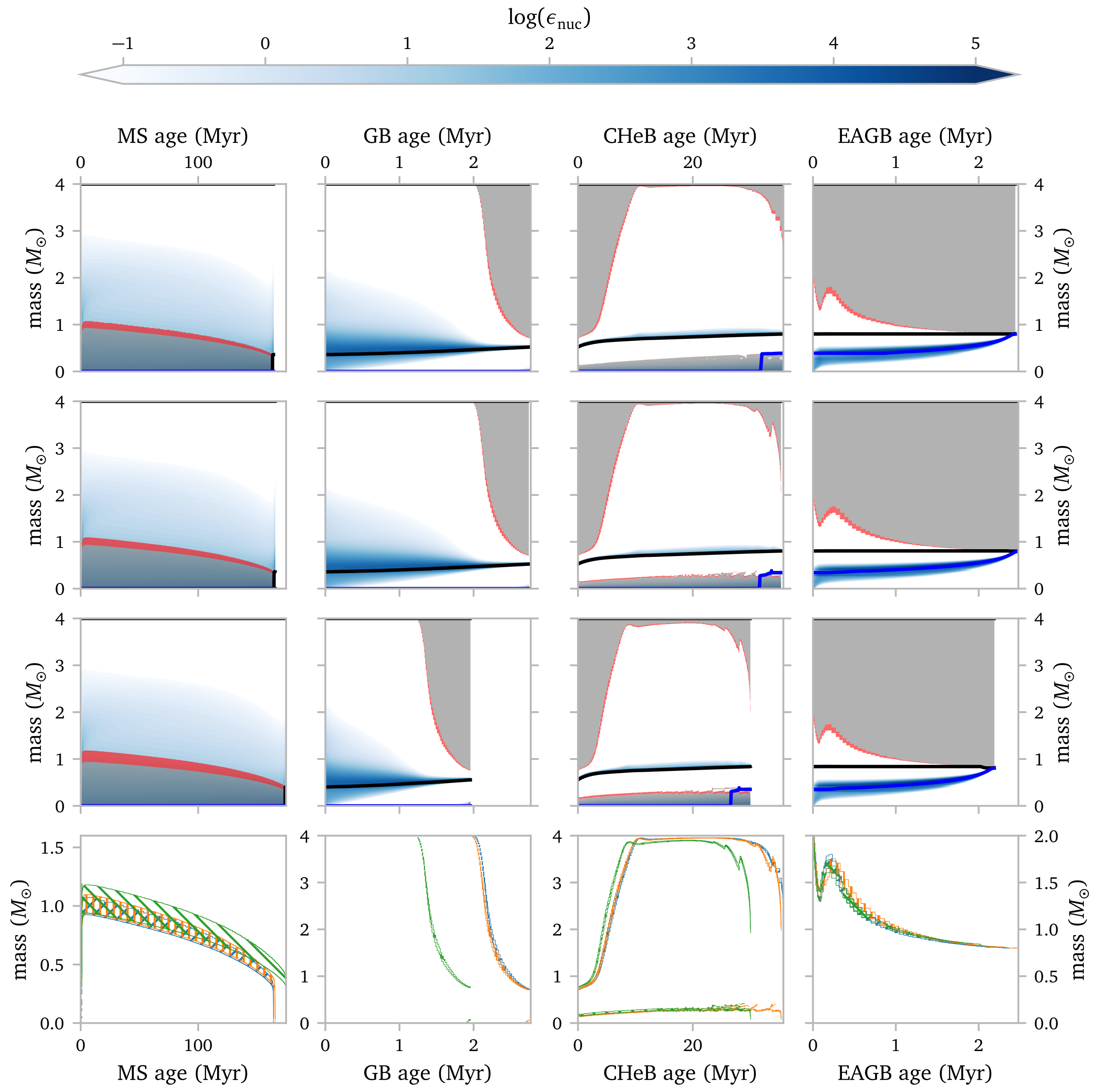
The tracks in the HR-diagrams for stars using the overshooting options from Rees et al. (2024) and Temmink et al. (2023) are very similar. When using the other $f_{\rm ov,0}$, the results are however significantly different. In general, these stars have higher luminosities.

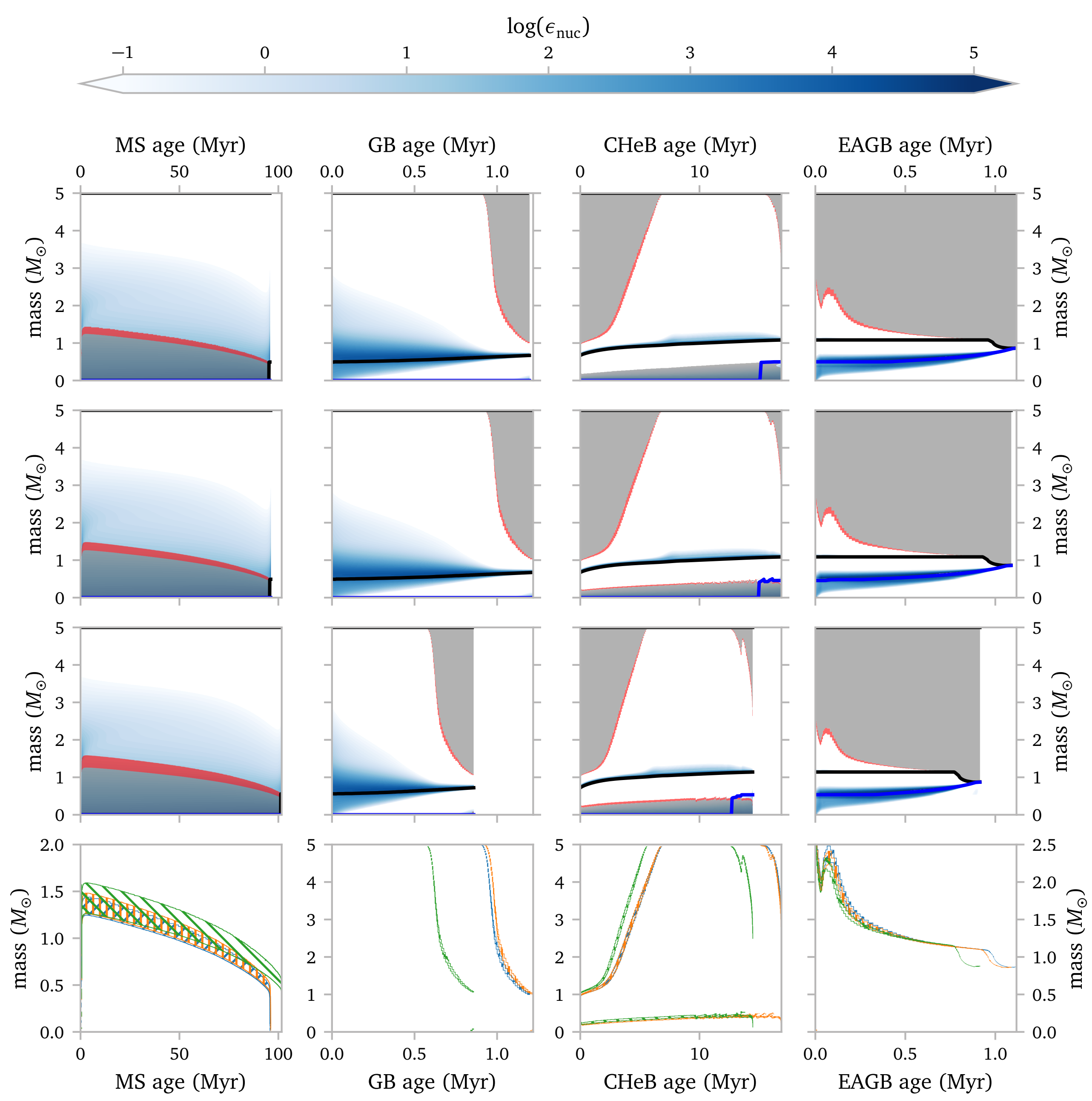
Below, I show Kippenhahn diagrams of the evolutionary stages for all stars. For completeness, I plot all of them, but really, the different masses are not very different.

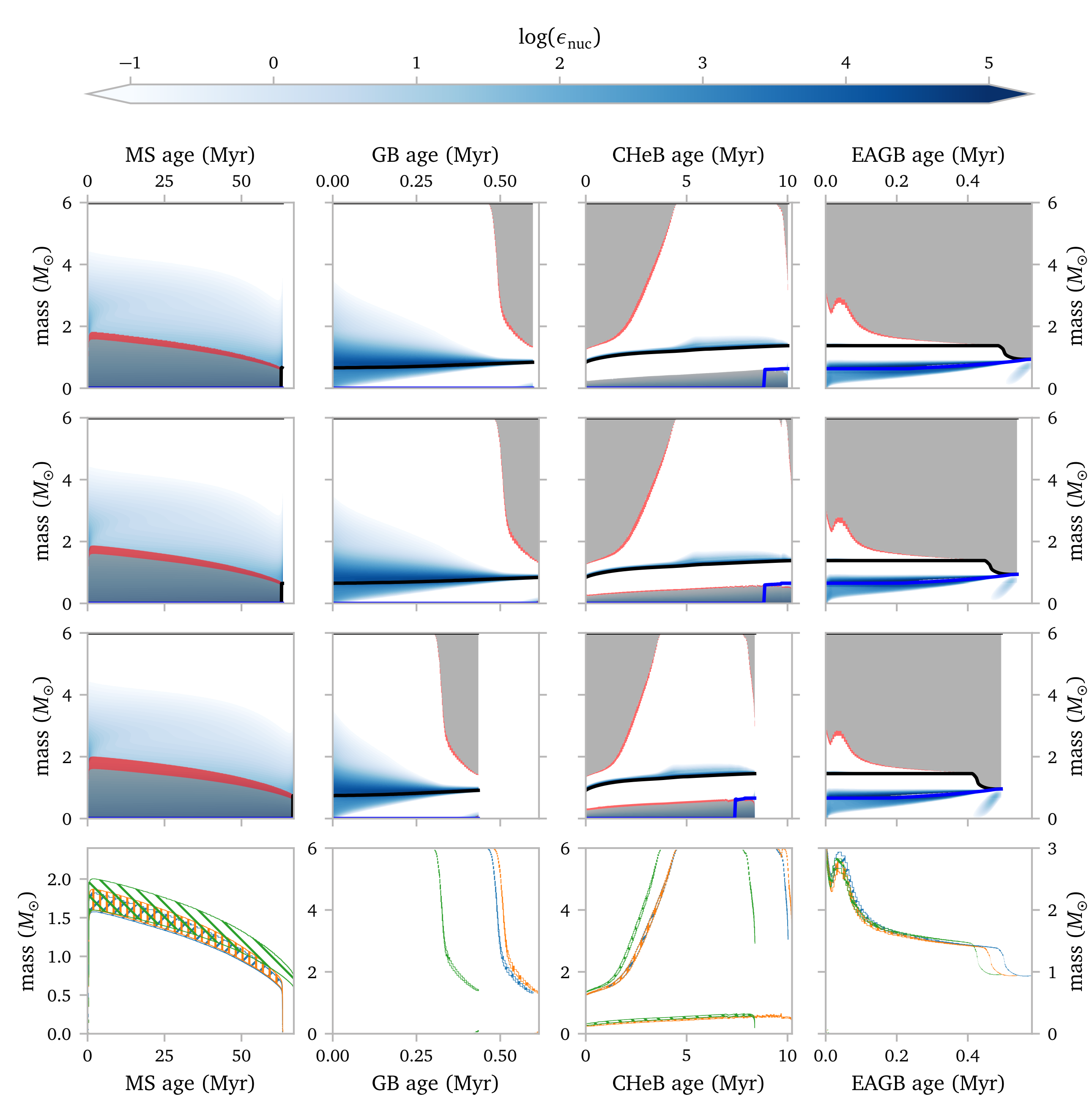
The gray regions are convective regions in the star. The red regions are overshooting regions. The blue "contours" show where nuclear burning is happening. The black line is the He-core mass and the blue line is the CO-core mass.

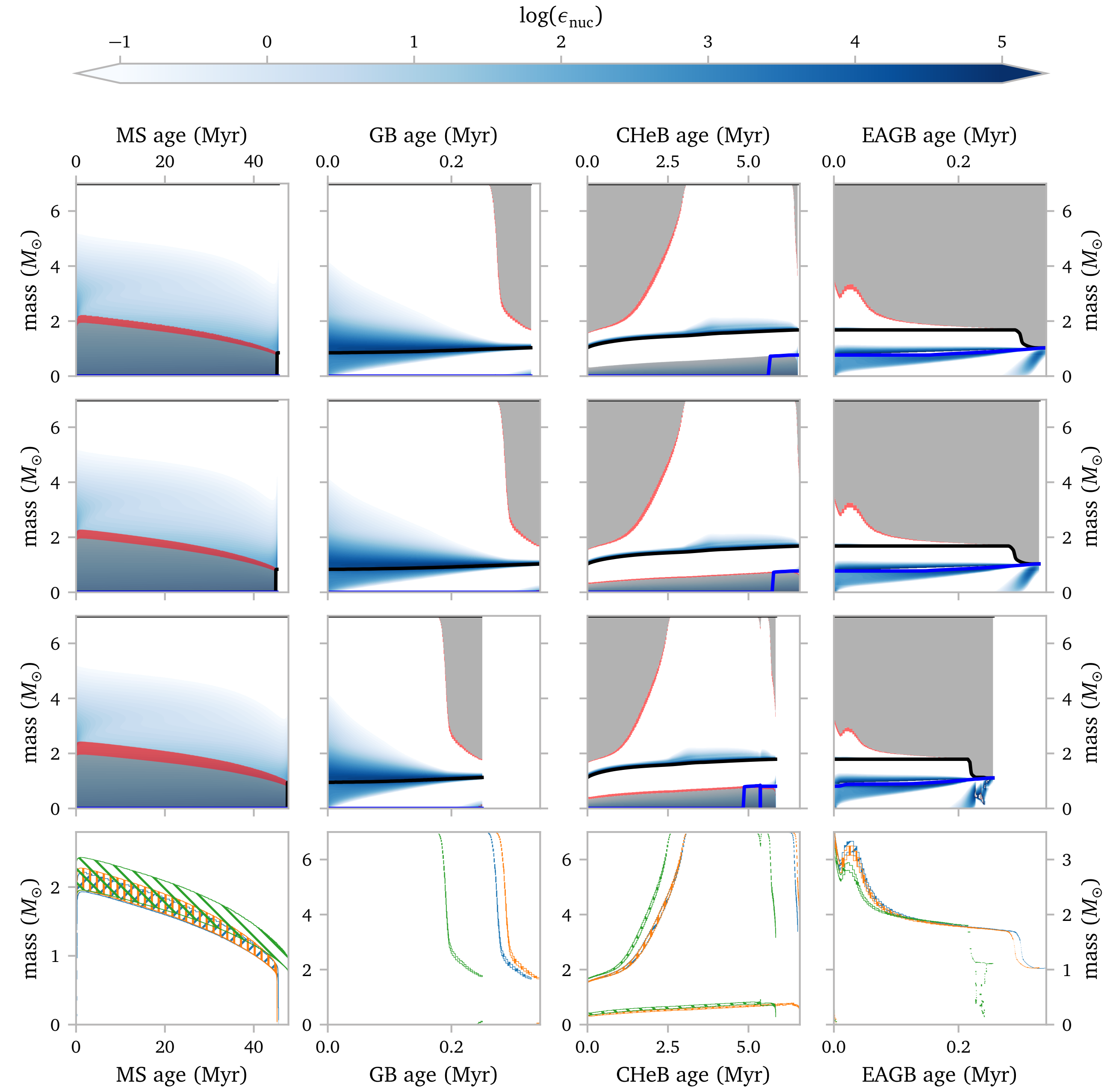


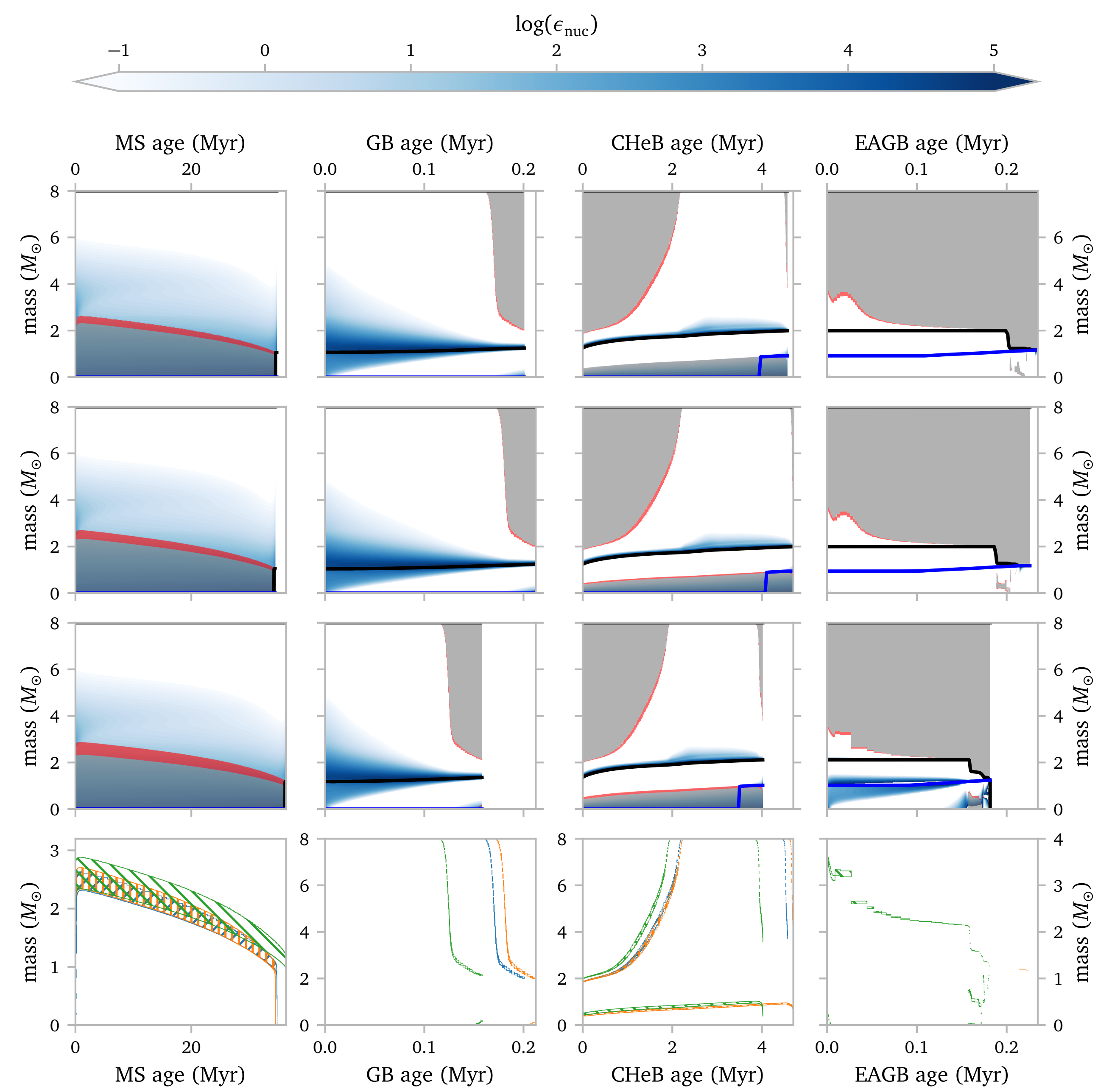



log(ϵnuc)
MS age (Myr)
GB age (Myr)
CHeB age (Myr)
EAGB age (Myr)
mass (M⊙)


log(ϵnuc)
MS age (Myr)
GB age (Myr)
CHeB age (Myr)
EAGB age (Myr)
mass (M⊙)


log(ϵnuc)
MS age (Myr)
GB age (Myr)
CHeB age (Myr)
EAGB age (Myr)
mass (M⊙)





The core mass at the start of the TPAGB is indistiguishable for the different models.

References

Rees, N. R., Izzard, R. G., & Karakas, A. I. (2024) *Thermal pulses with MESA: resolving the third dredge-up*. MNRAS527, 9643.

Temmink, K. D., Pols, O. R., Justham, S., Istrate, A. G., & Toonen, S. (2023) *Coping with loss. Stability of mass transfer from post-main-sequence donor stars*. A&A669, A45.